C8.2 Barrier transaction signaling

This section describes the read address channel and write address channel signaling associated with barrier transactions.

To permit interworking between barrier transactions and QoS, the following types of non-barrier transactions exist:

- transactions that are affected by barrier transactions
- transactions that are not affected by barrier transactions.

This specification recommends that, by default, all transactions are affected by barrier transactions. The only transaction streams that can be signaled so that they are not affected by barrier transactions are transactions that do not require ordering with respect to other streams, such as those related to real-time data flows.

C8.2.1 AxBAR signaling

AxBAR is used to differentiate between barrier transactions and normal transactions. For normal transactions **AxBAR** also indicates if the associated transaction must respect barriers or if the ordering requirements of any barrier transactions can be ignored. For barrier transactions, **AxBAR** also indicates whether the transaction is a memory barrier or a synchronization barrier. See *Read and write barrier transactions* on page C3-160 for more information about **AxBAR** encoding.

Table C3-13 on page C3-166 shows the constraints that apply to barrier transactions.

C8.2.2 AxDOMAIN signaling

The **AxDOMAIN** signal determines the level of propagation of a barrier transaction, defining the domains within a system that the barrier transaction accesses. See *Shareability domain types* on page C3-158. Table C8-1 shows the different levels of barrier applicability for each domain type.

Table C8-1 Domain barrier applicability

AxDOMAIN	Domain	Barrier Applicability
00	Non-shareable	Acts as a barrier to other transactions in the current transaction stream. When two or more transaction streams are combined, no ordering is required with respect to the newly combined transaction stream.
01	Inner Shareable	Ordering must be established with respect to other transactions from all masters in the same Inner Shareable domain.
10	Outer Shareable	Ordering must be established with respect to other transactions from all masters in the same Outer Shareable domain.
11	System	Ordering must be established with respect to all other transactions. For a Synchronization barrier, a response must only be given when all transactions from the issuing master, which are ahead of the barrier, have reached their endpoint.

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C8.2.3 Response signaling

All barrier transactions must complete with a response, as follows:

- responses for barrier transactions issued on the read address channel are signaled on the read data channel
- responses for barrier transactions issued on the write address channel are signaled on the write response channel.

The response must have a matching AXI ID to the barrier transaction and OKAY is the only permitted response for a barrier transaction. Table C8-2 shows the constraints for barrier transaction response signaling.

Table C8-2 Barrier response transaction constraints

Attribute	Constraint
RID, **BID**	Must match barrier transaction ID.
RRESP	Must be all zeros.
RLAST	Must be `0b01`.
RDATA	No constraint, can take any value. Must be ignored.
BRESP	Must be all zeros.

Note

User defined signals, such as **AxUSER**, **RUSER**, **WUSER**, and **BUSER**, cannot be reliably transported alongside barrier transactions. It is therefore recommended that user defined signals are all zeros for barrier transactions.

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